Cybersecurity Intrusion Detection Using Machine Learning

[Institution / University Name]
ECSCI24302 - [Course title as per syllabus]
Team: [Team member names]
Supervisor: [Supervisor / Course coordinator]
Date: [Allotted date]

Abstract

A leak-free supervised pipeline detects malicious flows in the UNSW-NB15 official partitioned release: 175,341 raw training rows (deduplicated to 107,740) and 82,332 untouched testing flows. With identical leakage-controlled pipelines, the selected HistGradientBoosting (tuned) achieves test accuracy 0.8712, precision 0.8154, recall 0.9903, F1 0.8944, ROC-AUC 0.9868 and average precision 0.9903 at a validation-chosen threshold of 0.480. Findings are reported honestly: 440 missed attacks (FNR 0.97%) but 10,164 false alarms (FPR 27.47%) - the release's train/test shift explains the FPR; the leakage sensitivity check (4,247 shared exact flows, 5.16%; F1 0.8944 vs 0.8902 unique-only) quantifies the inflation. Permuting sttl costs 0.3748 F1, but sttl is partly a schema artifact of the synthetic capture.

1. Introduction and problem definition

This project answers a single measurable question: on UNSW-NB15, how well can a leak-free supervised pipeline separate malicious from benign flows, and with what honest deployment caveat (the aim is to miss few attacks while reporting an honest false-alarm rate). Scope is an academic benchmark, NOT a production NIDS. All numbers quoted come verbatim from results/findings.md.

2. Background and related work

UNSW-NB15 (2015) mixes real Waterloo and UNSW captures with synthetically generated modern attacks; the referenced Moustafa & Slay papers document the design, the statistical properties and the train/test differences. Structured trees follow Breiman (CART, Random Forests); boosting follows Hastie et al.'s generic treatment.

3. Dataset and provenance

Item	Value
Name	UNSW-NB15 official partitioned release (UNSW Canberra Cyber, 2015)
Access page	https://www.kaggle.com/datasets/mrwellsdavid/unsw-nb15
Download mirror used	https://huggingface.co/datasets/Mireu-Lab/UNSW-NB15
Rows	175,341 training raw (dedup to 107,740) / 82,332 testing
Columns	45 per file (49 numeric + 3 categorical in the modeling schema; 16 engineered
Classes	Binary target; 10 attack families including Normal
Accessed	2026-10-26

4. Methodology and leakage-control rules

The official training partition is the only fitting source; the testing partition is held until a single final evaluation.

Data-dependent preprocessing is fitted on training data only and after split.

No model/hyper-parameter/threshold choice uses the test partition (one final predict only).

id, label, attack_cat are excluded from inputs (no target-derived predictor).

Every model shares one Pipeline so processing is identical.

All randomness is seeded (random_state=42) making the run deterministic.

5. Data quality audit

check	training	testing
rows	175,341	82,332
missing cells	0	0
exact duplicate rows (excl. id)	67,601	26,387
label/attack_cat mismatches	0	0

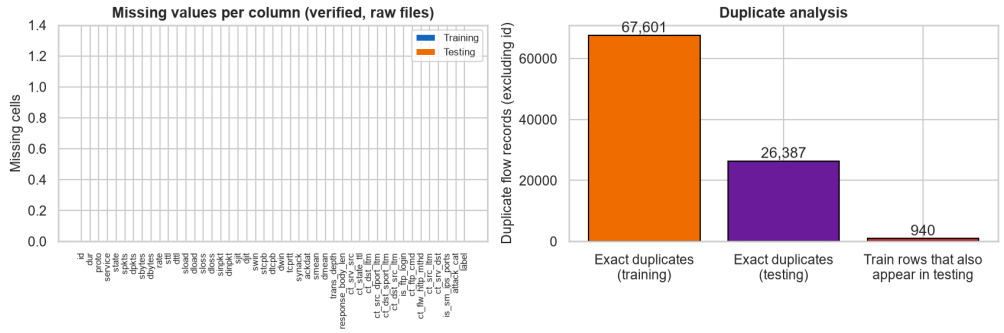


Figure 6: Missing cells and duplicate counts.

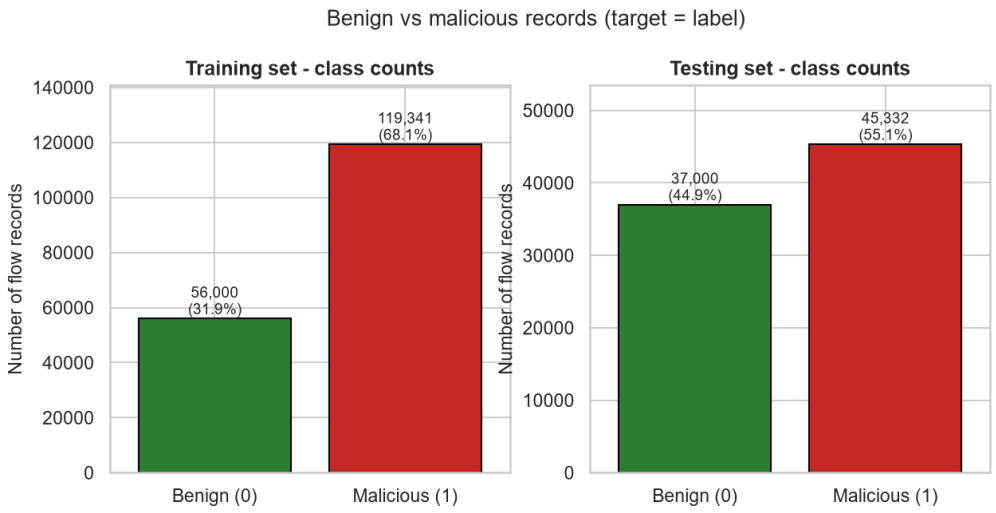


Figure 1: Class balance - benign vs malicious per partition.

6. Exploratory data analysis

Most positively correlated with 'malicious': sttl (0.693), ct_state_ttl (0.578), ct_dst_sport_ltm (0.357), rate (0.338), ct_src_dport_ltm (0.306), ct_dst_src_ltm (0.304), ct_src_ltm (0.238), ct_dst_ltm (0.230)

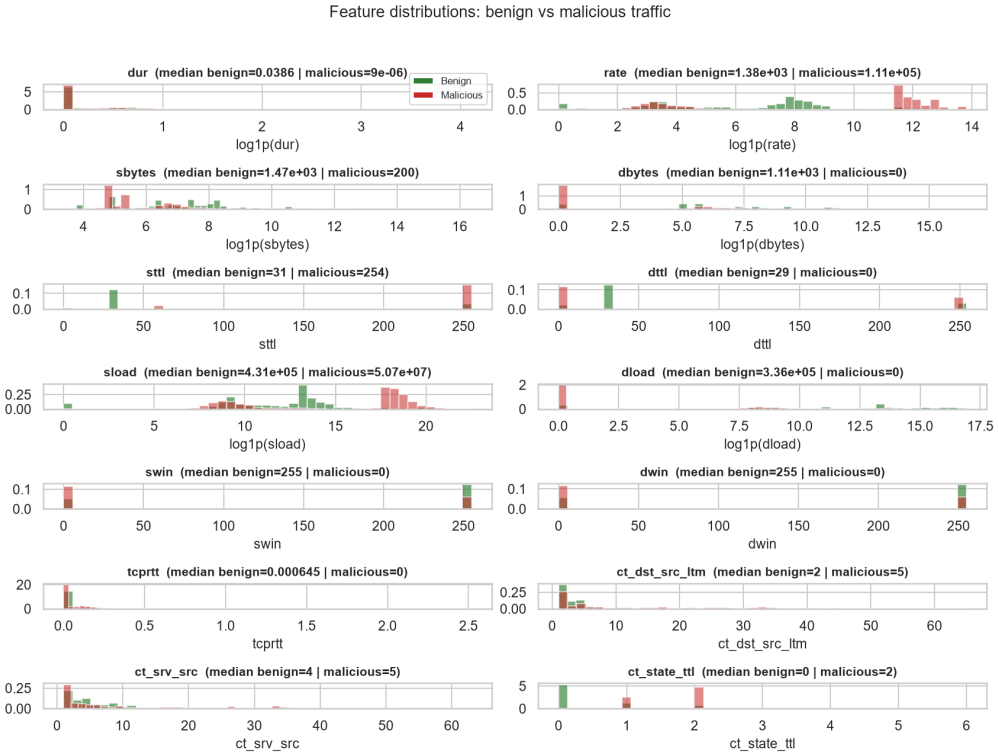


Figure 3: Key feature histograms by class.

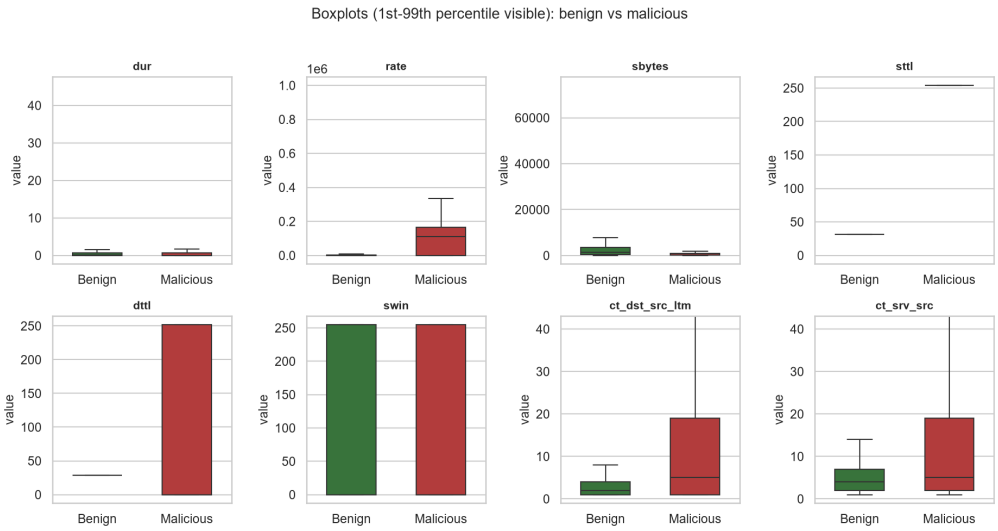


Figure 4: Box plots separating benign and malicious flows.

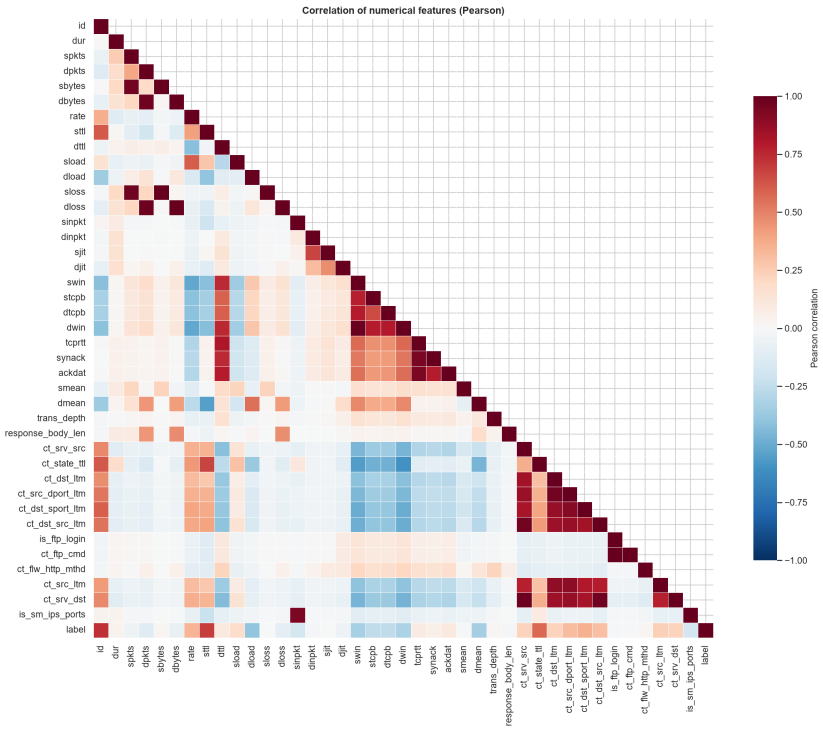


Figure 5: Feature correlation matrix.

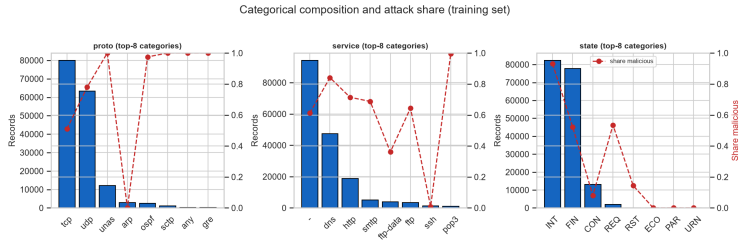


Figure 7: proto/service/state by attack share.

7. Model zoo and pipelines

Dummy (most_frequent) - floor baseline; never eligible for selection.

LogisticRegression (C=1.0, balanced, standardised).

DecisionTree (max_depth=12, balanced).

RandomForest (200 estimators, balanced_subsample).

HistGradientBoosting (max_iter=250 before tuning).

8. Hyper-parameter tuning

model	searcher	best hyper-parameters	best CV F1
logregression	GridSearchCV	{"C": 3.0}	0.9119
dtree	GridSearchCV	{"class_weight": null, "max_depth": 22, "min_samples_leaf": 1}	0.9202
forest	RandomizedSearchCV	{"n_estimators": 400, "min_samples_leaf": 2, "max_features": 0.3, "max_depth	0.9346
gb	RandomizedSearchCV	{"min_samples_leaf": 40, "max_leaf_nodes": 63, "max_iter": 300, "learning_ra	0.9359

cv=3, scoring F1, stratified 60,000-row subsample of the training partition; search spaces fixed a priori and printed in results/tuning_*.json.

9. Validation, selection and operating threshold

model	CV F1	std	accuracy	AUC
HistGradientBoosting (tuned)	0.9390	0.0013	0.9357	0.9872
RandomForest (tuned)	0.9382	0.0011	0.9344	0.9870
DecisionTree (tuned)	0.9268	0.0015	0.9227	0.9788
LogisticRegression (tuned)	0.9125	0.0009	0.9037	0.9695
Dummy	0.6828	0.0000	0.5184	0.5000

Selection rule (fixed a priori): highest mean CV F1 among tuned candidates; dummy never eligible. Selected: HistGradientBoosting (tuned) with CV F1 0.9390 +/- 0.0013. Threshold = 0.480 (max F1 on the stratified 20% internal validation slice of the training partition; never on test).

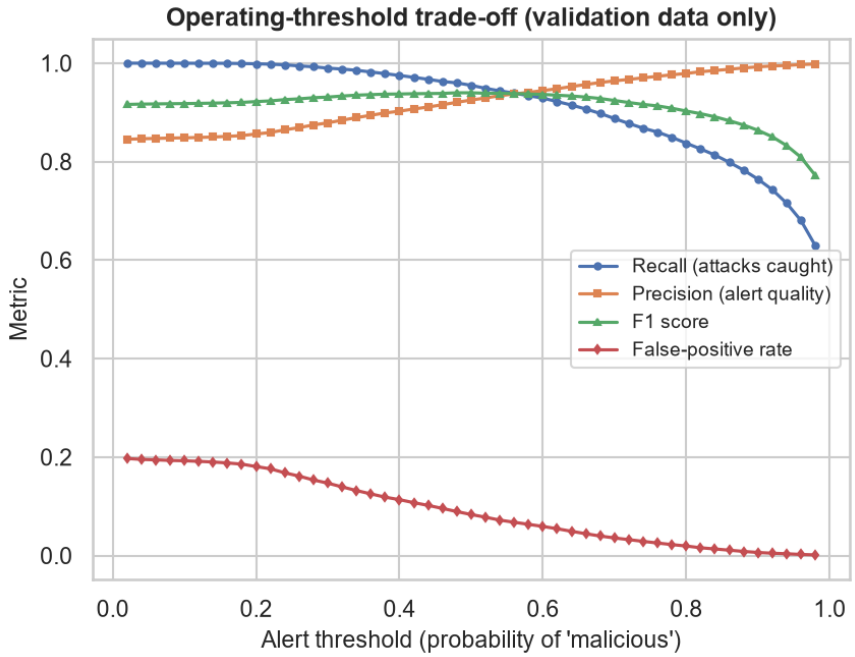


Figure 12: Operating-threshold trade-off (validation slice).

10. Evaluation on the untouched testing partition

model	accuracy	precision	recall	F1	AUC	AP
HistGradientBoosting (tuned)	0.8712	0.8154	0.9903	0.8944	0.9868	0.9903
RandomForest (tuned)	0.8663	0.8099	0.9894	0.8907	0.9841	0.9881
DecisionTree (tuned)	0.8601	0.8098	0.9749	0.8847	0.9693	0.9680
LogisticRegression (tuned)	0.8109	0.7550	0.9696	0.8490	0.9531	0.9650
Dummy	0.5506	0.5506	1.0000	0.7102	0.5000	0.5506

Chosen model: accuracy 0.8712, precision 0.8154, recall 0.9903, F1 0.8944, ROC-AUC 0.9868, AP 0.9903, FPR 27.47%, FNR 0.97%; confusion ben/mal: tn=26,836, fp=10,164, fn=440, tp=44,892.

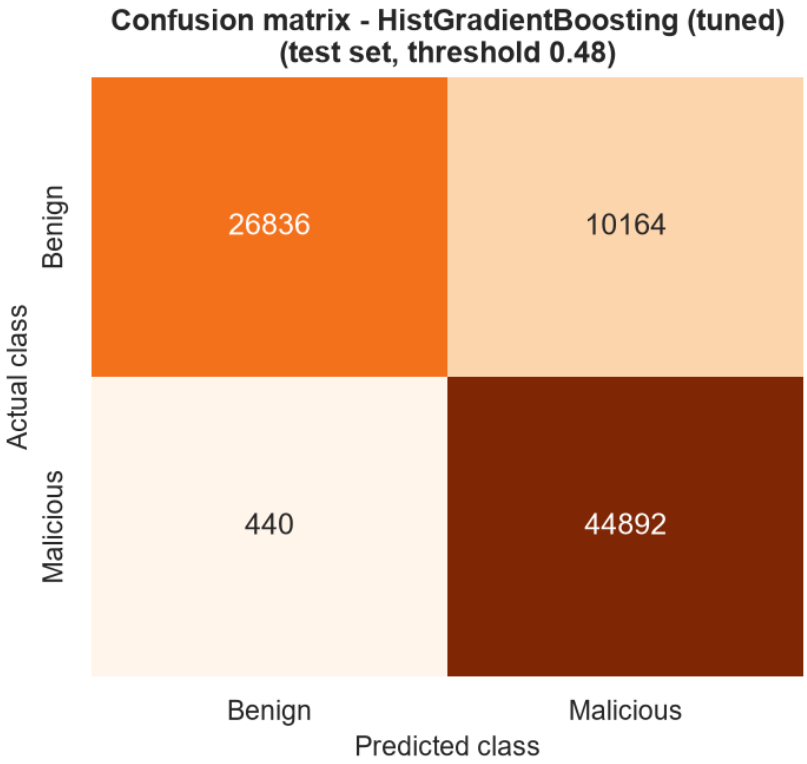


Figure 10: Confusion matrix of the chosen model (test).

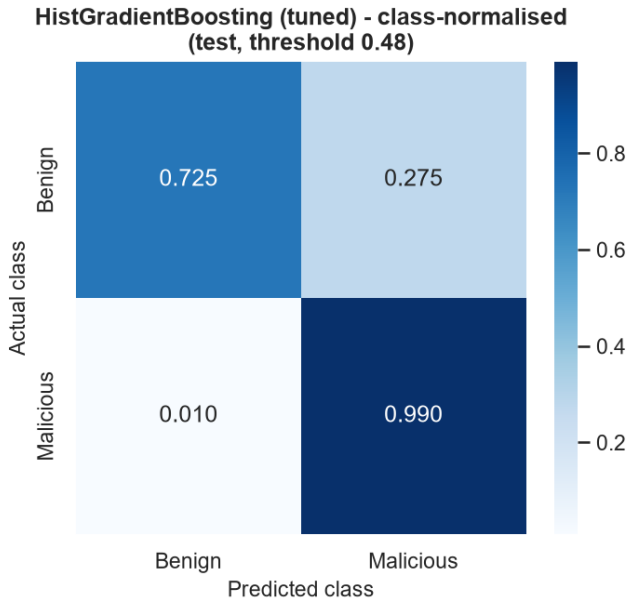


Figure 10b: Class-normalised confusion matrix.

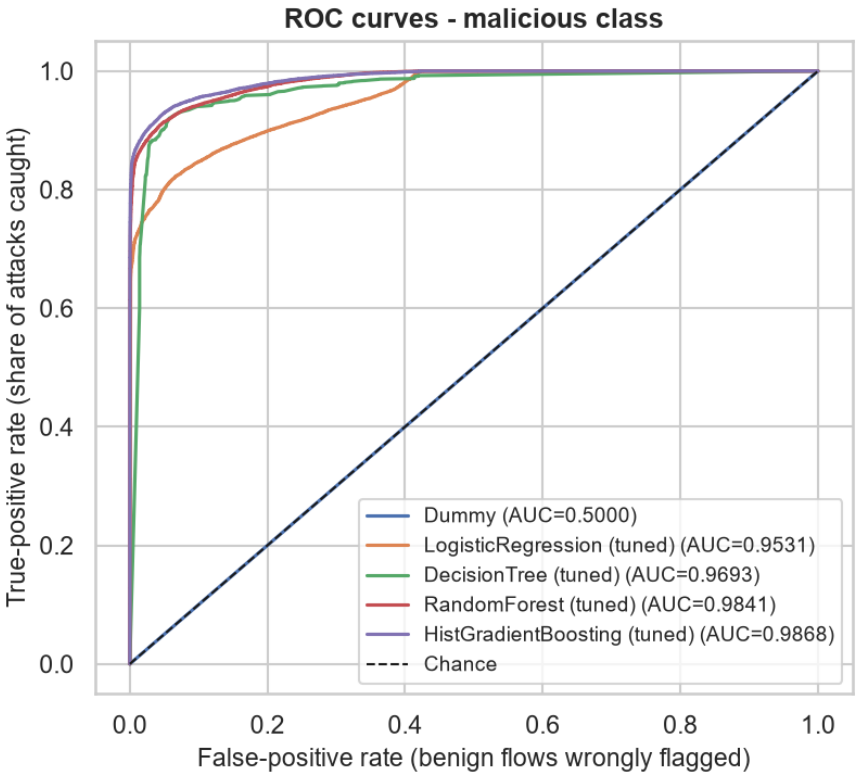


Figure 8: ROC curves on the untouched testing partition.

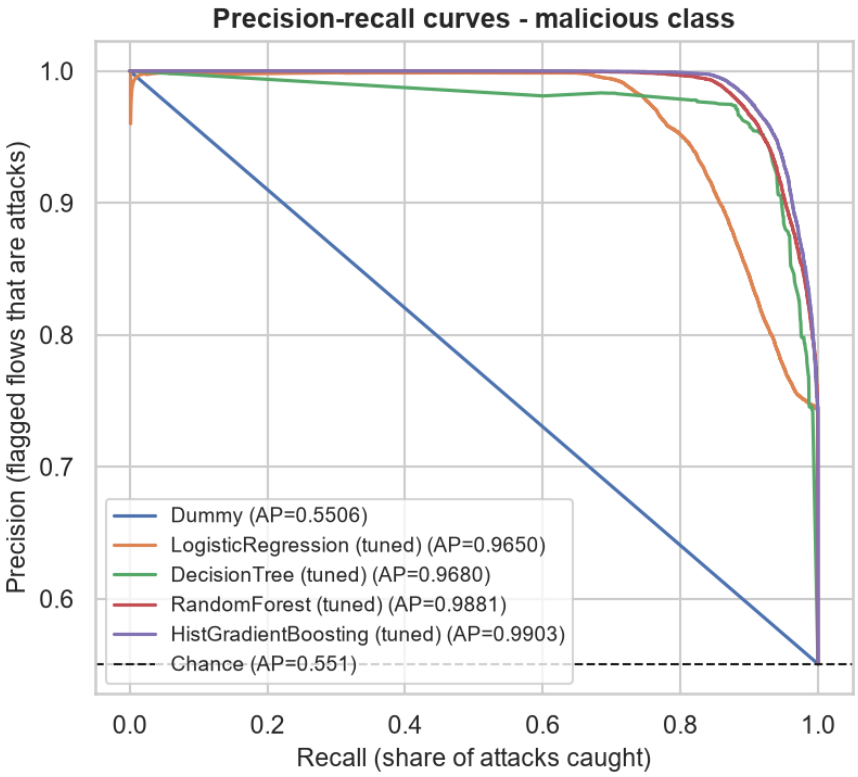


Figure 9: Precision-recall curves on the testing partition.

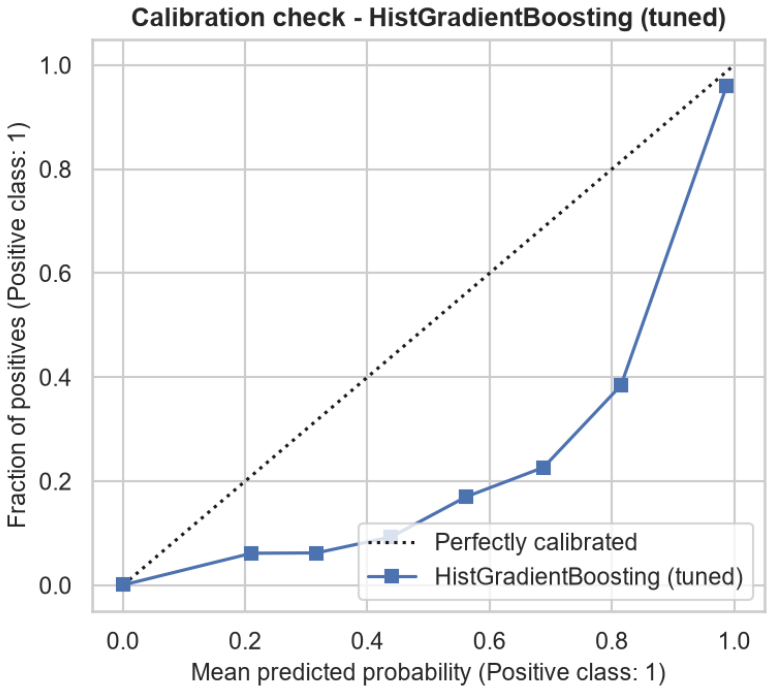


Figure 14: Calibration of the chosen model.

11. Error analysis

False positives: 10,164 (27.47% of benign) - FPs skew toward long-lived, high-TTL benign flows (FP medians: dur 0.676 vs 0.063 correctly-benign; rate 27.2 vs 654.5; sttl 254 vs 31). FP composition: 71% TCP / 29% udp; 71% FIN / 29% INT. False negatives: 440 (0.97% of attacks). Hardest: Fuzzers (372 of 6,062 missed, 6.14%); easiest: Backdoor (0 missed).

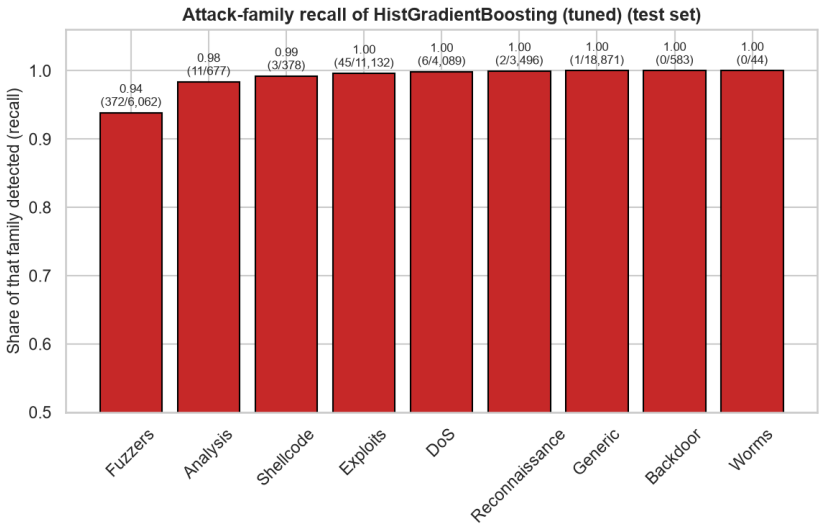


Figure 15: Recall by attack family (test).

12. Interpretation and caveats

Permutation importance on a stratified 10% sample of the training partition: sttl dominates (0.3748) - a known schema artifact of the synthetic capture; next features (dbytes 0.0228, sbytes 0.0202, ct_srv_dst 0.0174) are classic flow-shape signals. Importance is statistical, not causal: an important feature does NOT prove malicious intent.

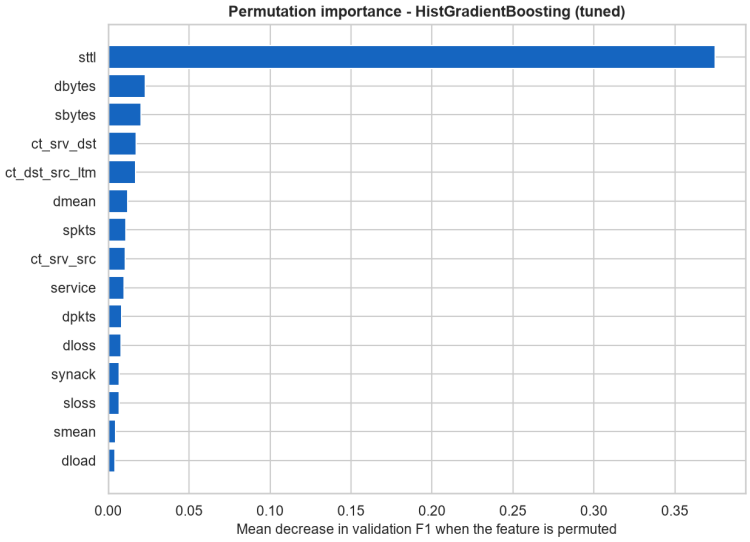


Figure 13: Permutation importance.

13. Robustness, leakage sensitivity and limitations

subset	accuracy	precision	recall	F1
all test rows	0.8712	0.8154	0.9903	0.8944
unique-only (78,085 rows)	0.8679	0.8090	0.9896	0.8902

FPR 27.47%: consequence of the release's train/test shift and the recall-favouring threshold; disclosed, not hidden.

sttl dominance (0.3748): synthetic-capture schema artifact; does not extrapolate beyond UNSW-NB15.

Multi-class macro-F1 0.5129: Analysis (recall 38.26%) and DoS (recall 12.40%) are genuinely hard.

Demo accuracy 80% on 10 held-out records: small-sample sanity check, not a validation claim.

14. Optional multi-class extension

Constrained RandomForest (150 trees, depth 20, leaf 2) on 10 families: accuracy 0.6814, macro-F1 0.5129, weighted-F1 0.7345.

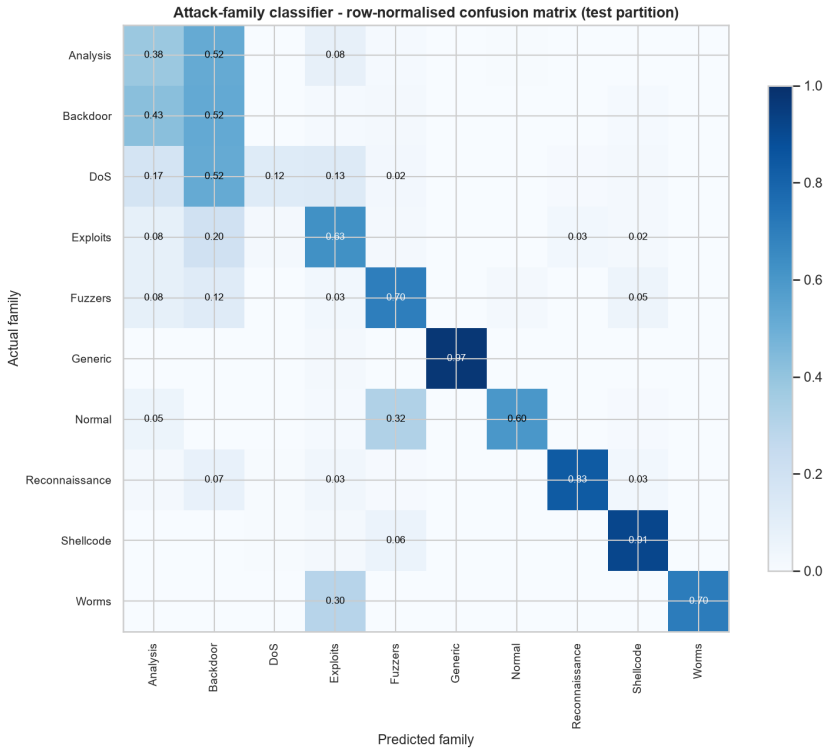


Figure 16: 10-family confusion matrix.

15. Conclusion

Under one strict leakage regime, HistGradientBoosting (tuned) delivers test F1 0.8944 with the honest caveat of a 27.47% false-alarm rate rooted in the release's own shift. Future work: distribution-aware thresholds, cost-sensitive selection, per-deployment calibration, external validation on other captures.

16. References

[1] N. Moustafa and J. Slay, "UNSW-NB15: a comprehensive data set for network intrusion detection systems." Military Communications and Information Systems Conference (MilCIS), 2015.

[2] N. Moustafa, G. Creech, and J. Slay, "Big data analytics for large scale wireless network security and attack detection." AFRICOMM, 2015.

[3] N. Moustafa and J. Slay, "The evaluation of network anomaly detection systems: statistical analysis of the UNSW-NB15 dataset and the comparison with the KDD99 dataset." Information Security Journal: A Global Perspective, 25(1-3):18-31, 2016.

[4] L. Breiman, "Random forests." Machine Learning, 45(1):5-32, 2001.

[5] L. Breiman, J. Friedman, R. Olshen, and C. Stone, "Classification and Regression Trees." Chapman & Hall/CRC, 1984.

[6] T. Hastie, R. Tibshirani, and J. Friedman, "The Elements of Statistical Learning: Data Mining, Inference, and Prediction," 2nd ed. Springer, 2009.

[7] F. Pedregosa et al., "Scikit-learn: machine learning in Python." Journal of Machine Learning Research, 12:2825-2830, 2011.

[8] W. McKinney, "Data structures for statistical computing in Python." SciPy 2010 (pandas).

Appendix - Reproduction instructions

pip install -r requirements.txt

python -m src.download_data

python -m src.train # ~6.2 min

jupyter lab notebooks/Cybersecurity_Intrusion_Detection.ipynb

streamlit run app/app.py

pytest

Honesty statement: every number in this PDF is verbatim from results/findings.md or sibling files produced by one pipeline run; unflattering results are disclosed, not hidden.